A Quick Review of OLS

WTF : A (Causal) relationship btw x_1 and y.

A simple reg model w/ one exp variable (x_1).

$y_i = \beta_0 + \beta_1 x_{1i} + u_i$ $\quad i$ = obs $\quad$ assuming $E(u_i|x_i) = 0$

ⓐ

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i$$
$$-)\ \bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \bar{u}$$
$$y_i - \bar{y} = \beta_1 (x_{1i} - \bar{x}_1) + (u_i - \bar{u})$$

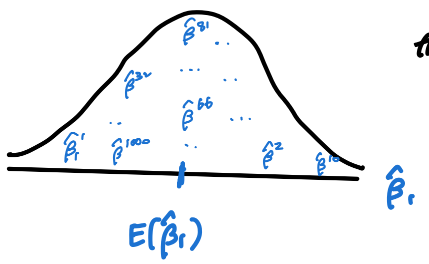
$$\hat{\beta}_1 = \frac{\Sigma(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\Sigma(x_{1i} - \bar{x}_1)^2} = \frac{cov(x_1, y)}{var(x_1)}$$

What does "bias" mean?

From random sampling with a large sample,

if we estimate β_1 multiple times, we have $\hat{\beta}_1^1, \hat{\beta}_1^2, \hat{\beta}_1^3 \ldots \hat{\beta}_1^{1000}$.
(1,000 times)

If we plot all these values and if it looks like this :



then $E(\hat{\beta}_1) = \beta_1$.

On average, estimated $\hat{\beta}_1$ equals the true parameter β_1.

In this case, we say that $\hat{\beta}_1$ is unbiased.

To sum up, we say that $\hat{\beta}_1$ is unbiased if $E(\hat{\beta}_1) = \beta_1$.

Is $\hat{\beta}_1$ from the OLS unbiased ?

$$\hat{\beta}_1 = \frac{\Sigma(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\Sigma(x_{1i} - \bar{x}_1)^2} \overset{ⓐ}{=} \frac{\Sigma(x_{1i} - \bar{x}_1)\left[\beta_1 (x_{1i} - \bar{x}_1) + (u_i - \bar{u})\right]}{\Sigma(x_{1i} - \bar{x}_1)^2}$$

$$= \beta_1 + \frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}$$

$$E(\hat{\beta}_1) = E\left[\beta_1 + \frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}\right]$$

$$= \beta_1 + \cancelto{\emptyset}{E\left[\frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}\right]}$$

because $E[u_i|x_i] = 0$

What if $E[u_i|x_i] \neq 0$?

Then, we have something other than β_1.

$\therefore \hat{\beta}_1$ is biased.

What is Omitted Variable Bias?

Suppose the true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

$$\text{where } E[u_i | x_{1i}, x_{2i}] = \phi$$

ⓑ

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$$
$$-)\ \bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \bar{u}$$
$$y_i - \bar{y} = \beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (u_i - \bar{u})$$

If we mistakenly run a regression w/o x_2,

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i \quad \text{, where } u_i = x_{2i} + \varepsilon_i$$

$$\tilde{\beta}_1 = \frac{\Sigma(x_{1i}-\bar{x}_1)(y_i-\bar{y})}{\Sigma(x_{1i}-\bar{x}_1)^2} \overset{ⓑ}{=} \frac{\Sigma(x_{1i}-\bar{x}_1)\left[\beta_1(x_{1i}-\bar{x}_1) + \beta_2(x_{2i}-\bar{x}_2) + (u_i-\bar{u})\right]}{\Sigma(x_{1i}-\bar{x}_1)^2}$$

$$= \beta_1 + \beta_2 \cdot \frac{\Sigma(x_{1i}-\bar{x}_1)(x_{2i}-\bar{x}_2)}{\Sigma(x_{1i}-\bar{x}_1)^2} + \frac{\Sigma(x_{1i}-\bar{x}_1)(u_i-\bar{u})}{\Sigma(x_{1i}-\bar{x}_1)^2}$$

$$E[\tilde{\beta}_1] = E\left[\beta_1 + \beta_2 \cdot \frac{\Sigma(x_{1i}-\bar{x}_1)(x_{2i}-\bar{x}_2)}{\Sigma(x_{1i}-\bar{x}_1)^2} + \frac{\Sigma(x_{1i}-\bar{x}_1)(u_i-\bar{u})}{\Sigma(x_{1i}-\bar{x}_1)^2}\right]$$

$$= \beta_1 + \beta_2 E\left[\frac{\Sigma(x_{1i}-\bar{x}_1)(x_{2i}-\bar{x}_2)}{\Sigma(x_{1i}-\bar{x}_1)^2}\right] + \cancel{E\left[\frac{\Sigma(x_{1i}-\bar{x}_1)(u_i-\bar{u})}{\Sigma(x_{1i}-\bar{x}_1)^2}\right]} \rightarrow \phi$$

$$= \beta_1 + \beta_2 \cdot \frac{cov(x_1, x_2)}{var(x_1)}$$

Isn't this familiar?

- regress y on x

$$y = a_0 + a_1 x + e$$

$$a_1 = \frac{cov(x, y)}{var(x)}$$

- what we have:

$$\frac{cov(x_1, x_2)}{var(x_1)}$$

$\Rightarrow$ It's the coeff on x_1 from regressing x_2 on x_1

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

$$= \beta_1 + \beta_2 \cdot \delta_1$$

Direct rel btw y & x_1

Direct rel tw x_1 & x_2

Direct rel btw y & x_2

$$= \beta_1 + \underbrace{\beta_2 \delta_1}_{\text{Bias}}$$

To sum up,

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \cdot \delta_1$$

- Definition of Unbiasedness

  if $\hat{\beta}_1$ is unbiased, $E(\hat{\beta}_1) = \beta_1$.

$$E(\tilde{\beta}_1) = \begin{cases} \beta_1 & \text{if } \beta_2\delta_1 = \emptyset \text{ ("unbiased")} \Leftrightarrow \beta_2 = \emptyset \text{ OR } \delta_1 = \emptyset \\ \beta_1 + \beta_2\delta_1 & \text{if } \beta_2\delta_1 \neq \emptyset \text{ ("biased")} \Leftrightarrow \beta_2 \neq \emptyset \text{ AND } \delta_1 \neq \emptyset \end{cases}$$

(∵) we call this term as "BIAS." because it biases our estimation.

What are β_2 and δ_1, again?

β_2 : Direct rel btw y & x_2 from $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$

δ_1 : Direct rel btw x_1 & x_2 from $x_{2i} = \delta_0 + \delta_1 x_{1i} + v_i$

Because this bias is caused by omitting x_2, which is related to BOTH y and x_1, we call this as Omitted Variable Bias (OVB).

Main Takeaway

If we run a regression omitting a variable that is BOTH correlated with y and x_1, our estimate has an OVB problem.
(We CANNOT say that x and y have a causal relationship.)

the variable of interest

(Example) LAB4. Question 7

True model : $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + \beta_3 \text{friend4} + \varepsilon$ (7)

What we estimated (in Q6) : $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + u$ (6)

<Auxiliary regression>

$$\text{Friend4} = \delta_0 + \delta_1 \text{indegree} + v$$
$$= \widehat{4.354} + \widehat{0.082}\ \text{indegree}$$

$$E(\tilde{\beta}_1) = \beta_1 + \underbrace{\underset{= 0.05}{\hat{\beta}_3} \cdot \underset{= 0.082}{\hat{\delta}_1}}_{\text{Bias} > 0}$$

Bias > 0 "upward bias"

the TRUE parameter β_1 from the true model (7)

What we estimated from Equation (6)

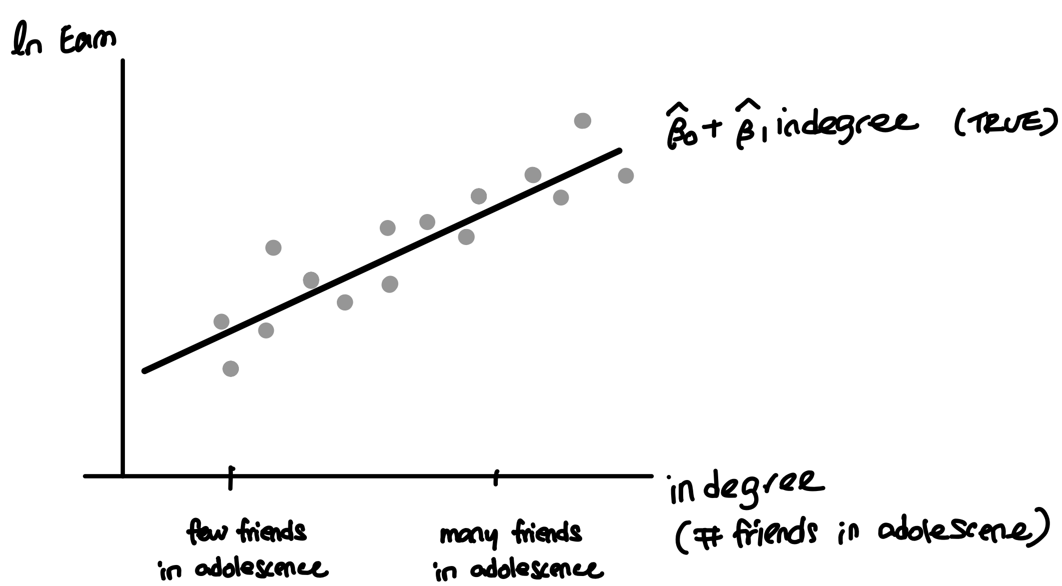
⇒ What can you say about $\tilde{\beta}_1$?

The estimated $\tilde{\beta}_1$ is larger than the true parameter β_1.

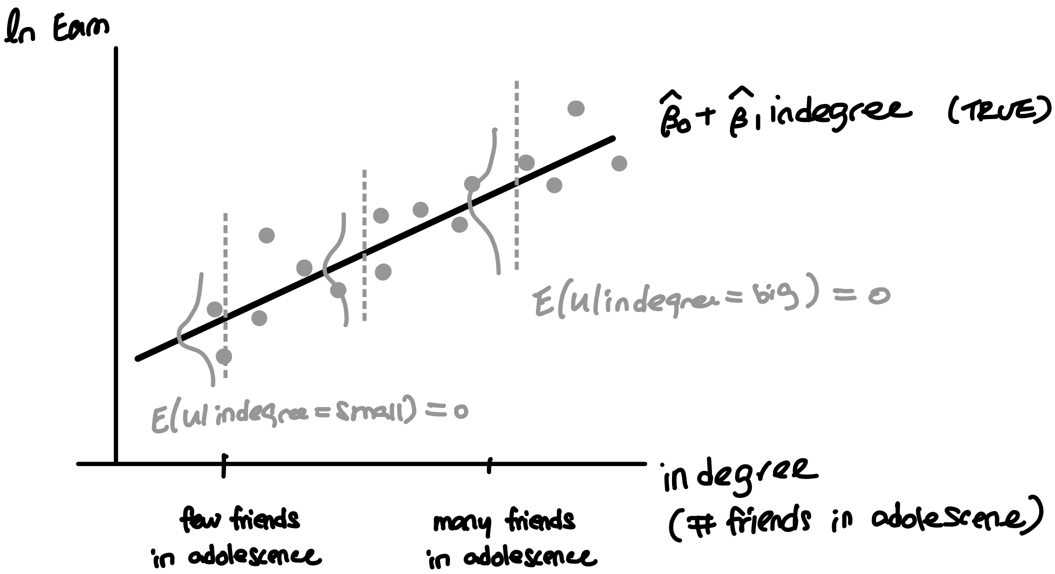
(i.e.) The true parameter is **smaller** than what we estimated in (6).

We also say that "$\tilde{\beta}_1$ is biased away from zero."

Suppose this is from the true model

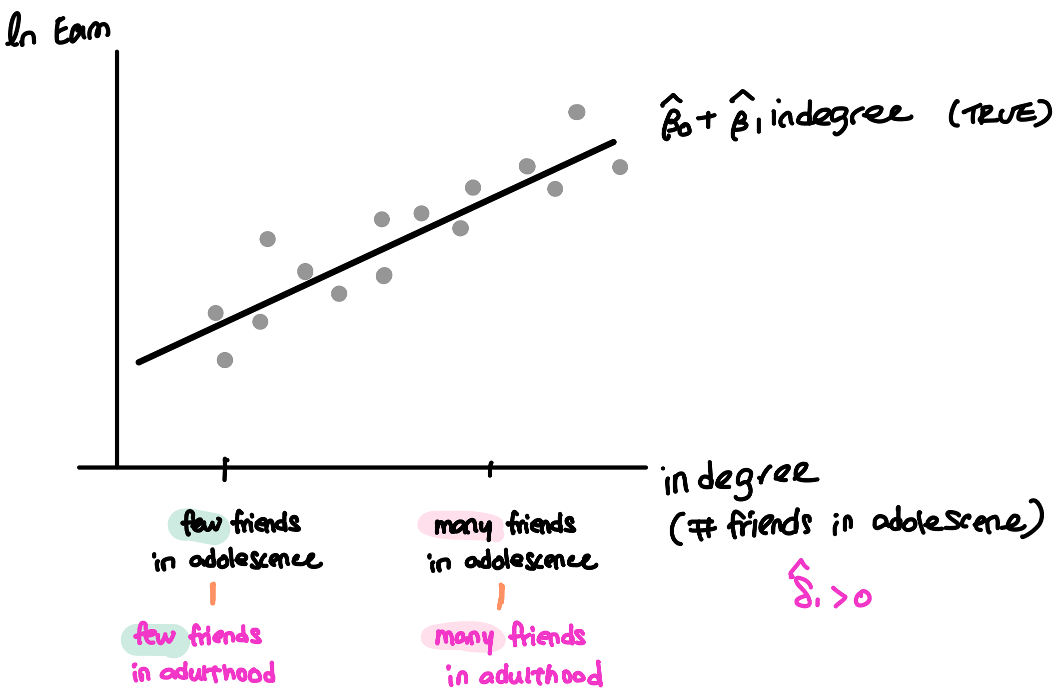


Suppose this is from the true model

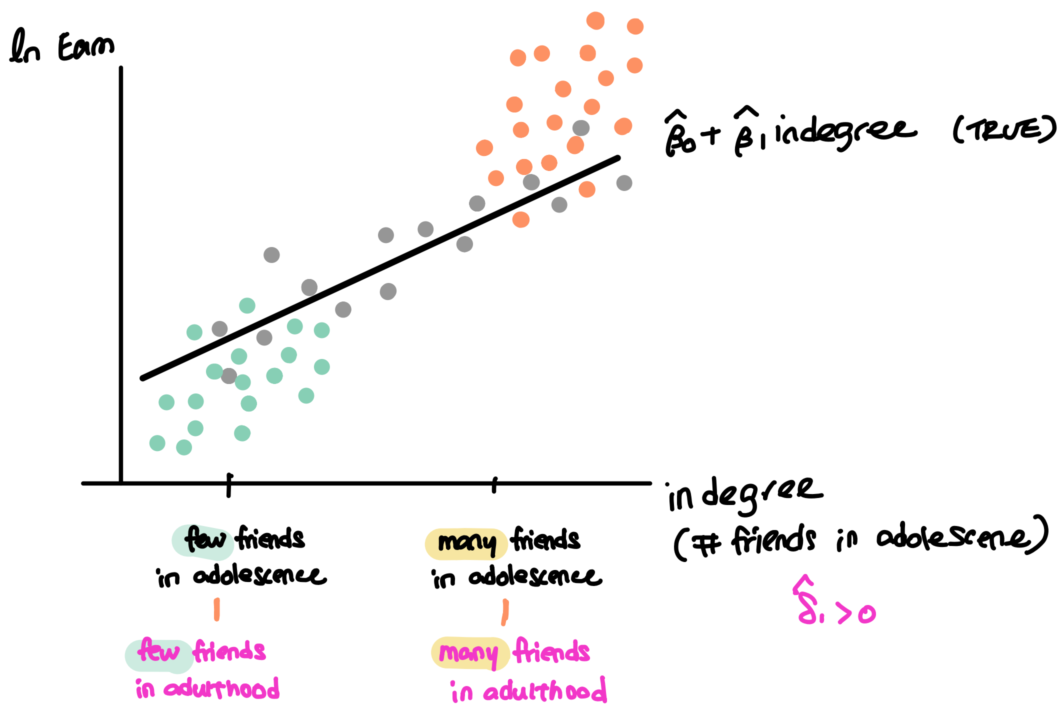


However, if we omit 'Friend_4' in the model, $\underbrace{\text{it}}_{u = \text{Friend\_4} + \varepsilon}$ is now part of the error term.

$\hat{\delta}_1 > 0$: "A positive rel btw # friends in adolescence & # friends in adulthood."

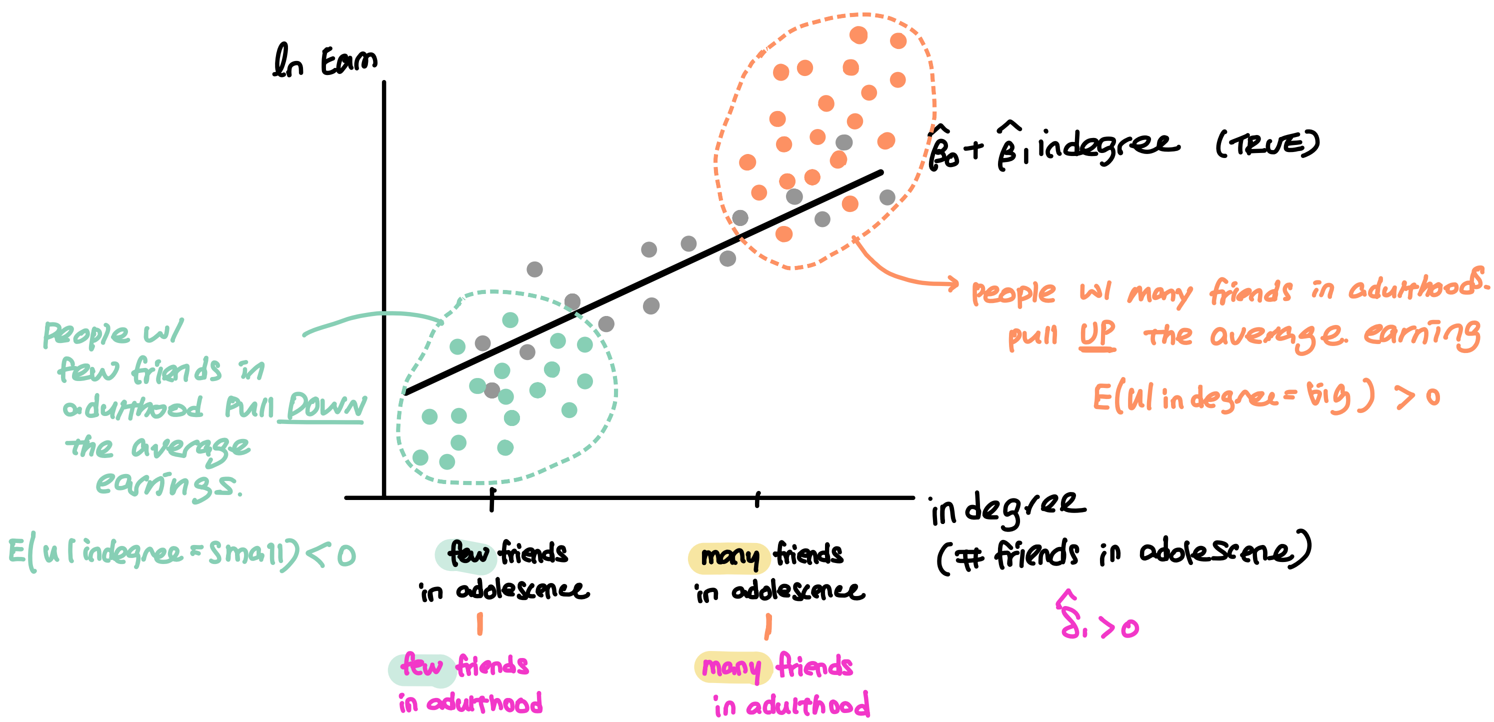


$\hat{\beta}_2 > 0$: "A positive relationship btw Earnings & # friends in adulthood."

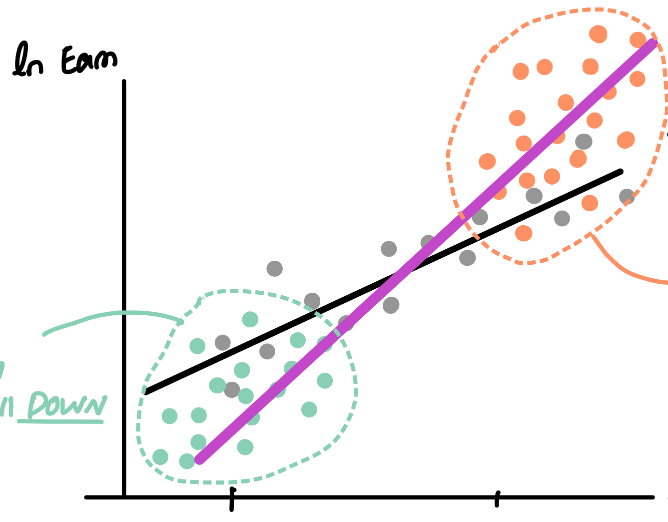


If we omit Friend_4 in the reg model, the error term u = Friend_4 + ε.

Because Friend_4 is correlated with lnEarn AND in_degree, $E[u \mid \text{in\_degree}] \neq \phi$.



Therefore ...



$\tilde{\beta}_0 + \tilde{\beta}_1$ indegree (Omitted)

$\hat{\beta}_0 + \hat{\beta}_1$ indegree (TRUE)

People w/ few friends in adulthood pull DOWN the average earnings.

$E(u \mid indegree = small) < 0$

People w/ many friends in adulthood pull UP the average earnings.

$E(u \mid indegree = big) > 0$

indegree (# friends in adolescence)

few friends in adolescence | few friends in adulthood

many friends in adolescence | many friends in adulthood

$\hat{\delta}_1 > 0$

True model : $\ln Earn = \beta_0 + \beta_1 indegree + \beta_2 female + \beta_3 friend4 + \varepsilon$

```
. reg ln_earn in_degree female friend_4
```

Source	SS	df	MS
Model	134.501255	3	44.8337517
Residual	2705.73506	2,597	1.04186949
Total	2840.23631	2,600	1.09239858

Number of obs	=	2,601
F(3, 2597)	=	43.03
Prob > F	=	0.0000
R-squared	=	0.0474
Adj R-squared	=	0.0463
Root MSE	=	1.0207

ln_earn	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
$\hat{\beta}_1$ in_degree	.0200568	.005484	3.66	0.000	.0093035	.0308102
female	-.3045655	.0404073	-7.54	0.000	-.3837993	-.2253316
$\hat{\beta}_3$ friend_4	.0503541	.0073086	6.89	0.000	.0360229	.0646853
_cons	10.00184	.0513622	194.73	0.000	9.901127	10.10256

```
. reg friend_4 in_degree
```

: $Friend4 = \delta_0 + \delta_1 indegree + v$ (Auxiliary regression)

Source	SS	df	MS
Model	268.635594	1	268.635594
Residual	22103.3174	2,891	7.64556119
Total	22371.953	2,892	7.7358067

Number of obs	=	2,893
F(1, 2891)	=	35.14
Prob > F	=	0.0000
R-squared	=	0.0120
Adj R-squared	=	0.0117
Root MSE	=	2.7651

friend_4	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
$\hat{\delta}_1$ in_degree	.0824532	.0139101	5.93	0.000	.0551785	.1097279
_cons	4.353976	.0828171	52.57	0.000	4.191589	4.516362

ECON 108 Discussion Section

Week 2. Simple Linear Regression Model: Practice Questions

$Y_i = \beta_0 + \beta_1 X_i + u_i$

Question 1.

$\text{Test score}_i = \beta_0 + \beta_1 STR_i + u_i$

Suppose we are interested in examining the effect of class size, measured by the student-teacher ratio (str), on average test scores (testscr). The following information is based on school district-level data. X Y

```
. su testscr str
```

Variable	Obs	Mean	Std. dev.	Min	Max
Y testscr	420	654.1565 $\bar{Y}$	19.05335 S_Y	605.55	706.75
X str	420	19.64043 $\bar{X}$	1.891812 S_X	14	25.8

```
. corr testscr str, cov
(obs=420)
```

	testscr	str
Y testscr	363.03 → S_Y^2	
X str	-8.15932 cov(X, Y)	3.57895 → S_X^2

	X_1	X_2
X_1	$cov(X_1, X_1) = V(X_1)$	
X_2	$cov(X_1, X_2)$	$cov(X_2, X_2) = V(X_2)$

Let's calculate the estimated coefficients manually.

a. What is $\widehat{\beta_1}$? $\widehat{\beta_0}$? Interpret $\widehat{\beta_1}$.

$$\hat{\beta}_1 = \frac{cov(X, Y)}{S_X^2} = \frac{-8.159}{3.79} = -2.28$$

When a school district's STR increases by one student per class, test scores decrease by 2.28 points.

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \cdot \bar{X} = 654.157 - (-2.28)(19.64)$$
$$= 698.936$$

b. What is R^2? Interpret the meaning of the number.

$$R^2 = r^2 = (-0.226)^2 = 0.051$$

$$r = \frac{cov(X, Y)}{S_X \cdot S_Y} = \frac{-8.159}{(1.892)(19.053)} = -0.226$$

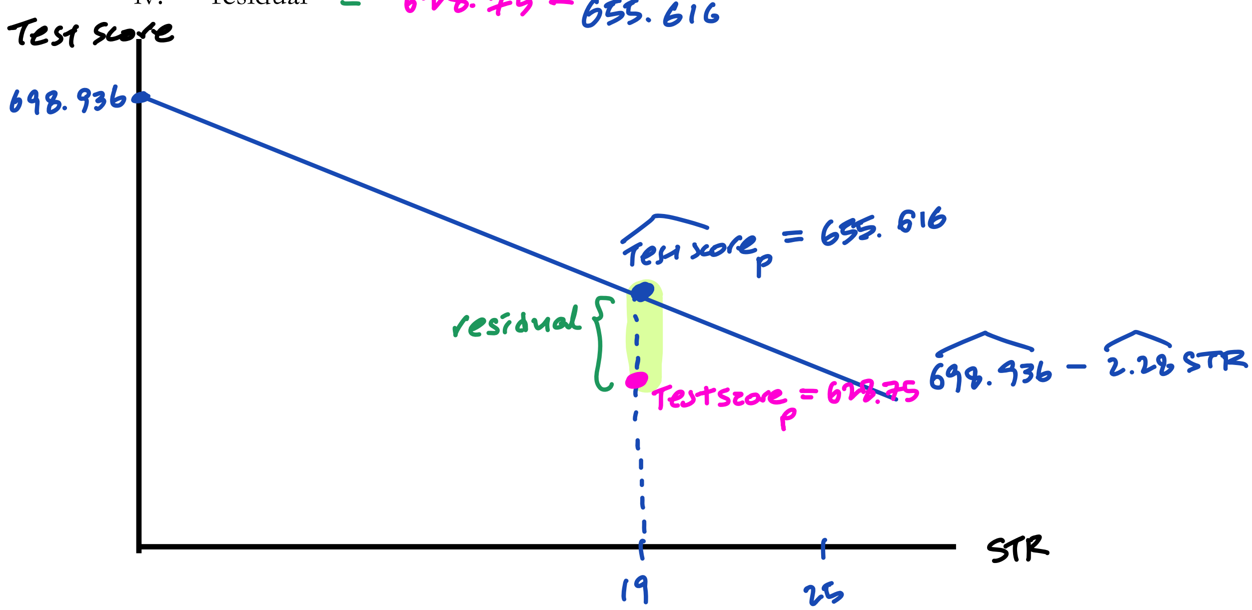
c. Predict testscr if the student-teacher ratio (str) is 19.

STR=19

$$\widehat{\text{Test score}}_i = \hat{\beta}_0 + \hat{\beta}_1 STR_i$$
$$= 698.936 - 2.28\, STR_i \quad =19$$
$$= 698.936 - 2.28(19)$$
$$= 655.616$$

d. The student-teacher ratio for Perris School District is 19, and the average test score is 628.75. In the graph below, draw the estimated regression line and label the following for Perris School District:

i. STR
ii. Actual average test score
iii. Predicted average test score
iv. residual $= 628.75 - \widehat{655.616}$



Did we calculate everything correctly?

```
. reg testscr str

      Source |       SS           df       MS      Number of obs   =       420
-------------+----------------------------------   F(1, 418)       =     22.58
       Model |  7794.11004         1  7794.11004   Prob > F        =    0.0000
    Residual |  144315.484       418  345.252353   R-squared       =    0.0512
-------------+----------------------------------   Adj R-squared   =    0.0490
       Total |  152109.594       419  363.030056   Root MSE        =    18.581

------------------------------------------------------------------------------
     testscr | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -2.279808   .4798256    -4.75   0.000     -3.22298   -1.336637
       _cons |    698.933   9.467491    73.82   0.000     680.3231    717.5428
------------------------------------------------------------------------------
```

$\hat{\beta}_1$ $\hat{\beta}_0$

```
. margins, at(str=19)

Adjusted predictions                                     Number of obs = 420
Model VCE: OLS

Expression: Linear prediction, predict()
At: str = 19

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _cons |   655.6166   .9573181   684.85   0.000     653.7348    657.4984
------------------------------------------------------------------------------
```

Question 2.

We are also interested in the relationship between the number of computers and average test scores. The regression results are presented below, with standard errors shown in parentheses.

$$\widehat{TestScore_i} = \underset{(1.127)}{\overset{\hat{\beta}_0}{655.122}} - \underset{(0.002)}{\overset{\hat{\beta}_1}{0.003}}\, Computer_i \qquad (1) \checkmark$$

$$\widehat{TestScore_i} = \underset{(2.08)}{643.36} - \underset{(13.81)}{79.40}\, Computer\ per\ Student_i \qquad (2)$$

a. Interpret the estimated coefficients. (Choose one model.)

Slope $\hat{\beta}_1$: Additional computer in a school district decreases test scores by 0.003, all else equal.

intercept $\hat{\beta}_0$: when a school district has zero computers, predicted test scores are 655.122.

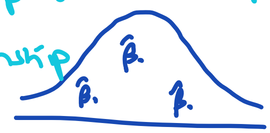
Ho

Ha

b. Conduct a significance test at the 5% level to evaluate whether there is any relationship between the number of computers and average test scores. Show all your steps. Be sure to show how you calculate the necessary probabilities from a z table.

① $H_0: \beta_1 = 0 \equiv \beta_{1,0}$ ← no relationship btw # comp & test scores

$H_a: \beta_1 \neq 0$ ← there is a relationship



② sig lv (α) = 0.05

③ test statistic ~~z-score~~ t $= \dfrac{\hat{\beta}_1 - \boxed{\text{mean of } \hat{\beta}_1 \text{ under Ho}}^{\beta_{1,0}}}{\text{sd of } \hat{\beta}_1 \text{ under Ho}}$

• mean of $\hat{\beta}_1$
$= E(\hat{\beta}_1) = \beta_1$

under Ho, $\beta_{1,0}$

• ~~sd~~ se of $\hat{\beta}_1 \rightarrow se(\hat{\beta}_1)$

$$= \frac{\hat{\beta}_1 - \beta_{1,0}}{se(\hat{\beta}_1)}$$

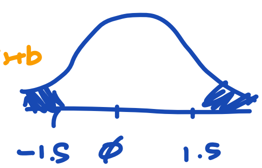
$$= \frac{\hat{\beta}_1 - \phi}{se(\hat{\beta}_1)}$$

$$= \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{-0.003}{0.002} = -1.5$$

④ p-value $\Rightarrow Pr(|t| > |\text{test statistic}|)$

$= Pr(|t| > 1.5)$

t distb ≈ z distb when sample size is large!



$= 2 \times \underbrace{P(\cancel{t}\,z < -1.5)}_{0.0668}$ z-table!

$= 0.1336$

⑤ Conclusion

P-value	vs	sig lv (α)
0.1336	>	0.05

⇒ fail to reject Ho

(no relationship btw # computers and test scores.)

c. Construct a 95% confidence interval for the slope parameter. Discuss how the results are complementary to your findings in part b).

* not $\hat{\beta}_1$

A 95% CI for β_1 $= \hat{\beta}_1 \pm 1.96 \cdot se(\hat{\beta}_1)$

95% of the time β_1 is w/in this interval

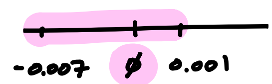
$$= -0.003 \pm \underbrace{1.96\,(0.002)}_{0.004}$$

$$= -0.003 \pm 0.004$$

$$= [-0.003 - 0.004\,,\; -0.003 + 0.004]$$

$$= [-0.007\,,\; 0.001]$$

contains $\emptyset$



∴ We cannot rule out that $\underbrace{\beta_1 \text{ is zero.}}_{\text{no relationship btw \# computers and test scores}}$

The actual regression results are shown below. How accurate were our manual calculations of the test statistic, p-value, and confidence interval (CI)?

```
. reg testscr computer

      Source |       SS           df       MS      Number of obs   =       420
-------------+----------------------------------   F(1, 418)       =      2.29
       Model |  827.015554         1  827.015554   Prob > F        =    0.1314
    Residual |  151282.578       418  361.920043   R-squared       =    0.0054
-------------+----------------------------------   Adj R-squared   =    0.0031
       Total |  152109.594       419  363.030056   Root MSE        =    19.024

------------------------------------------------------------------------------
     testscr | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    computer |  -.0031833   .0021058    -1.51   0.131    -.0073226    .0009561
       _cons |   655.1223   1.126888   581.36   0.000     652.9072    657.3374
------------------------------------------------------------------------------
```

test stat
Qb③

p-value
Qb④

Q3.

```
. reg testscr comp_stu

      Source |       SS           df       MS      Number of obs   =       420
-------------+----------------------------------   F(1, 418)       =     33.05
       Model |  11146.6436         1  11146.6436   Prob > F        =    0.0000
    Residual |   140962.95       418  337.231938   R-squared       =    0.0733
-------------+----------------------------------   Adj R-squared   =    0.0711
       Total |  152109.594       419  363.030056   Root MSE        =    18.364

------------------------------------------------------------------------------
     testscr | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    comp_stu |   79.40485   13.81145     5.75   0.000      52.2563    106.5534
       _cons |   643.3633   2.080229   309.28   0.000     639.2743    647.4523
------------------------------------------------------------------------------
```

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

ECON 108 Discussion Section

slope

Week 3. Multiple Linear Regression Model & Nonlinear Models

$y = \beta_0 + \beta_1 x + u \rightarrow \boxed{\frac{\Delta y}{\Delta x}} = \beta_1$

- **Agenda for week 3**: <u>Expand a simple linear regression model</u> by adding additional variables and incorporating nonlinear functions for the dependent and/or explanatory variables.

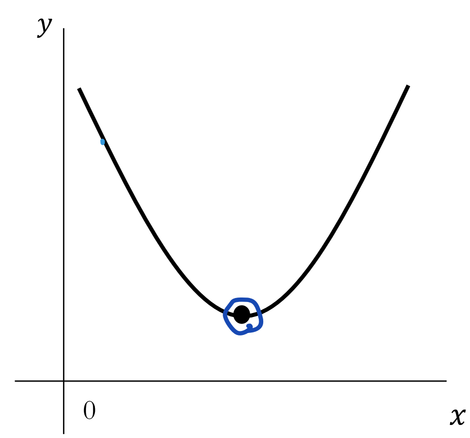
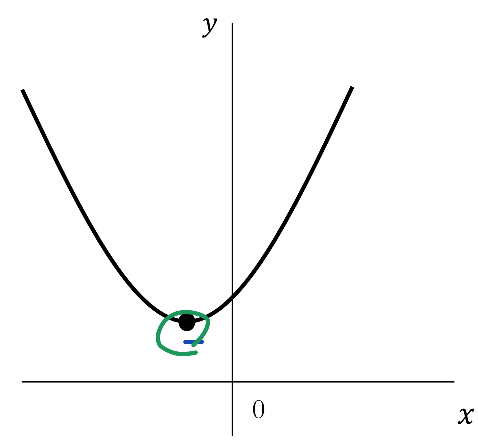
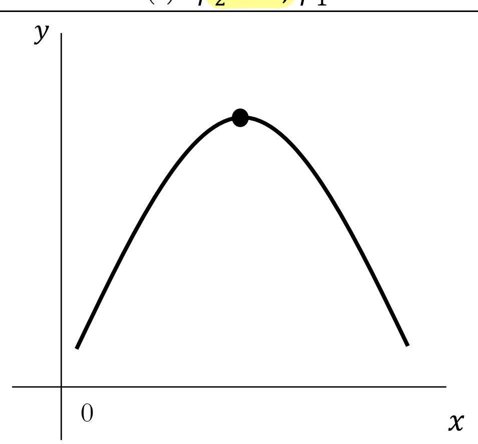
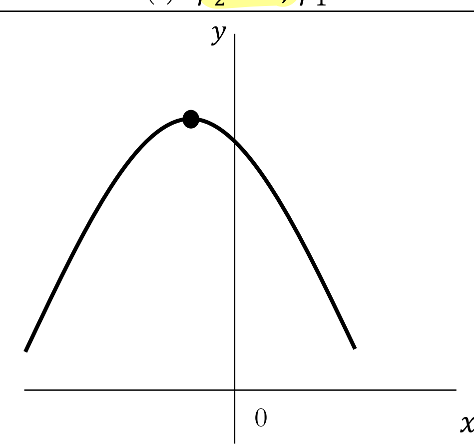
- **Shapes of a Quadratic Model**: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ $\quad \underbrace{\frac{\Delta y}{\Delta x}}_{\text{slope}} = \frac{dy}{dx} = \underbrace{\beta_1 + 2\beta_2 x}$

  The turnaround point? $\frac{\Delta y}{\Delta x} = 0 \Rightarrow \beta_1 + 2\beta_2 x = 0 \rightarrow 2\beta_2 x = -\beta_1$

  $\boxed{x^* = -\frac{\beta_1}{2\beta_2}}$

  How to tell the shape of the model based on the signs of coefficients? (Or vice versa.)

  1. Check the sign of coefficient for x^2 (β_2)
     a. $\beta_2 > 0$: a U-shaped curve (1) and (2)
     b. $\beta_2 < 0$: an inverted U-shape curve (3) and (4)
  2. Check the coefficient for x (β_1):
     a. If β_1 and β_2 have the same sign, the turning point is negative. (2) and (4)
     b. If β_1 and β_2 have the opposite signs, the turning point is positive. (1) and (3)

	
(1) $\beta_2 > 0,\ \beta_1 < 0$	(2) $\beta_2 > 0,\ \beta_1 > 0$
	
(3) $\beta_2 < 0,\ \beta_1 > 0$	(4) $\beta_2 < 0,\ \beta_1 < 0$

- **Testing the nonlinear relationship** between y and x in a quadratic model:

$$y = \beta_0 + \beta_1 x + \boldsymbol{\beta_2 x^2} + u$$

$$\boxed{y = \beta_0 + \beta_1 x + u}$$

H_0: $\beta_2 = 0$ (i.e., no need for a quadratic term)

H_a: $\beta_2 \neq 0$ (i.e., the quadratic term is necessary)

- **Interpreting nonlinear regression models**

Case	Regression	Interpretation	
lin-lin	$y_i = \beta_0 + \beta_1 x_i + u_i$ $\Delta y = \beta_1 \underbrace{\boxed{\Delta x}}_{1}$	A change in x by one unit is associated with a change in y by β_1 (unit), on average.	x: unit of x y: unit of y
lin-log	$y_i = \beta_0 + \beta_1 lnx_i + u_i$	A 1% change in x is associated with $0.01{*}\beta_1$ change in y, on average.	x: % y: unit
log-lin	$lny_i = \beta_0 + \beta_1 x_i + u_i$	A change in x by one unit is associated with a $100{*}\beta_1$% change in y, on average.	x: unit of x y: %
log-log	$lny_i = \beta_0 + \beta_1 lnx_i + u_i$	A 1% change in x is associated with a β_1% change in y, on average. β_1 is the elasticity of y with respect to X.	x: % y: %

Useful fact

$\Delta \ln x \times 100 = \% \Delta x$

lin-log

$$\underbrace{\Delta y}^{\times 100} = \beta_1 \cdot \underbrace{\Delta \ln x}^{\times 100}$$

$$\Delta y \cdot 100 = \beta_1 \cdot \% \Delta x$$

$$\Delta y = \left(\frac{\beta_1}{100}\right) \underbrace{\% \boxed{\Delta x}}_{1}$$

log-lin

$$\underbrace{\Delta \ln y}^{\times 100} = \beta_1 \Delta x^{\times 100}$$

$$\% \Delta y = \underbrace{(\beta_1 \times 100)}\, \underbrace{\boxed{\Delta x}}_{1}$$

log-log

$$\underbrace{\Delta \ln y}^{\times 100} = \beta_1 \cdot \underbrace{\Delta \ln x}^{\times 100}$$

$$\% \Delta y = \beta_1 \cdot \% \underbrace{\boxed{\Delta x}}_{1}$$

Question 1.

Consider the following regression model that examines the relationship between the number of bedrooms and housing price:

$$Price_i = \beta_0 + \beta_1 Room_i + \beta_2 Room_i^2 + u_i$$

a. What is the marginal effect of an additional bedroom on housing price?

$$\frac{\Delta Price}{\Delta Room} = \frac{d\,Price}{d\,Room} = \beta_1 + 2\beta_2 \cdot Room$$

b. What is the predicted marginal effect of an additional bedroom on housing price for a house with 2 bedrooms? = Room

$= 2$

$$\frac{\Delta Price}{\Delta Room} = \beta_1 + 2\beta_2 \cdot \boxed{Room}$$

$$= \beta_1 + 4\beta_2$$

c. Suppose we run an alternative model as follows. Interpret the coefficient on $ln(Size_i)$. The unit of Size is square feet.

log-log $\quad ln(Price_i) = \alpha_0 + \alpha_1 ln(Size_i) + \alpha_2 Room_i + u_i.$

If the size of a house increases by 1%, the housing price changes by α_1 %.

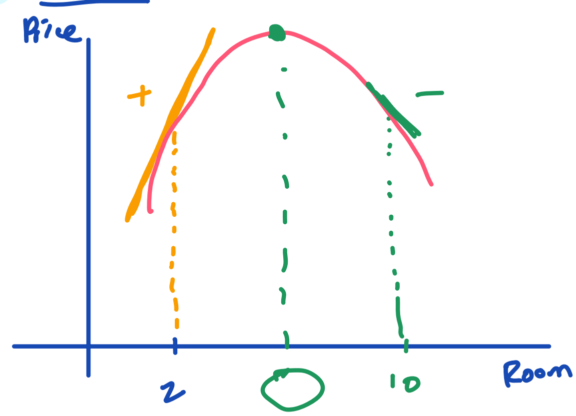
Question 2.

For the following scenario, predict the signs of the coefficients. Trace out the shape of the relationship between the dependent variable and the explanatory variable. $Price = \beta_0 + \beta_1 Room + \beta_2 Room^2 + u$

slope → The change in housing price from adding one room is \$50,000 when the current house has 2 bedrooms. The change in housing price from adding one room is -\$10,000 when the current house has 10 bedrooms.

$$\underbrace{\frac{\Delta Price}{\Delta Room}}_{slope} = \frac{d\,Price}{d\,Room} = \beta_1 + 2\beta_2 Room$$

slope $\begin{cases} 50{,}000 & \text{at } Room = 2 \\ -10{,}000 & \text{at } Room = 10 \end{cases}$



① $\beta_2 < 0$ b/c the slope changes from (+) to (−)

② $\beta_1 > 0$ b/c the turning pt is a positive number (somewhere btw 2 & 10), and $\beta_2 < 0$. For the turning pt to be positive, β_1 & β_2 must have opposite signs.

Question 3.

Our research on examining the relationship between the number of computers per student and average test scores continues this week. This time, we include the computer-student ratio (comp_stu) and its squared term (comp_stu2) to investigate whether the relationship between the computer-student ratio and test scores (testscr) is nonlinear. Additionally, the model controls for the student-teacher ratio (str). Use the following Stata results to answer the questions.

```
. gen comp_stu2 = comp_stu^2

. reg testscr comp_stu comp_stu2 str
```

$Score_i = \beta_0 + \beta_1 CSR_i + \beta_2 CSR_i^2 + \beta_3 STR_i + u_i$

Source	SS	df	MS
Model	15164.5307	3	5054.84358
Residual	136945.063	416	329.194863
Total	152109.594	419	363.030056

Number of obs	=	420
F(3, 416)	=	15.36
Prob > F	=	0.0000
R-squared	=	0.0997
Adj R-squared	=	0.0932
Root MSE	=	18.144

testscr	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
comp_stu	123.1777 $\hat\beta_1$	46.28552	2.66	0.008	32.19501	214.1603
comp_stu2	-173.3259 $\hat\beta_2$	131.4739	-1.32	0.188	-431.7619	85.11018
str	-1.533944 $\hat\beta_3$	.4943405	-3.10	0.002	-2.505661	-.5622273
_cons	671.4727	11.16598	60.14	0.000	649.5239	693.4214

a. Interpret the coefficient of the student-teacher ratio (str).

$\hat\beta_3 = -1.53$ "An add'l student per class decreases test scores by 1.53 points, all else equal."

b. The computer-student ratio in the Anaheim Elementary School District is $\boxed{0.05}$. What is the predicted change in test scores if the district purchases one additional computer per student?

$$\frac{\Delta Score}{\Delta CSR} = \frac{\partial Score}{\partial CSR} = \hat\beta_1 + 2\hat\beta_2 \overset{0.05}{CSR}$$

$$= 123.18 + 2 \cdot (-173.33)(0.05)$$

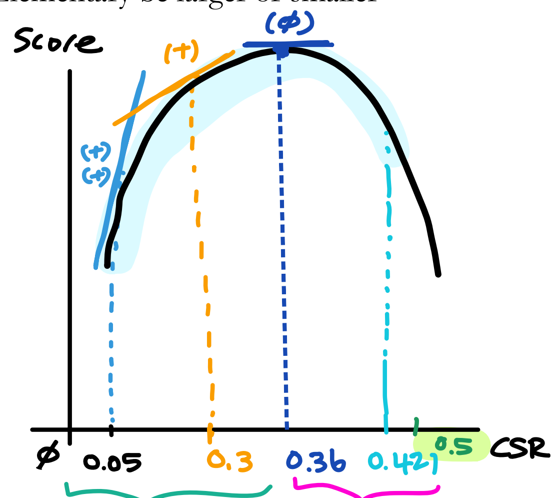
$$= 105.85$$

c. The computer-student ratio in the Portola Valley Elementary School District is 0.3. What is the predicted change in test scores if the district purchases one additional computer per student? Will the change in test scores for Portola Valley Elementary be larger or smaller compared to your answer in part b?

$$\frac{\Delta \text{Score}}{\Delta \text{CSR}} = \hat{\beta}_1 + 2\hat{\beta}_2 \cdot \text{CSR}$$

$$= 123.18 + 2(-173.33)(0.3)$$

$$= 19.182$$



add'l comp ↑ test scores

add'l comp ↓ test score

d. Find the turning point in the relationship between the computer-student ratio and test scores.

Turning pt ⇒ $\frac{\Delta \text{score}}{\Delta \text{CSR}} = \emptyset$

$$\therefore \hat{\beta}_1 + 2\hat{\beta}_2 \cdot \text{CSR} = \emptyset$$

$$\therefore \text{CSR}^* = -\frac{\hat{\beta}_1}{2\hat{\beta}_2} = \frac{-123.18}{2 \cdot (-173.33)} = 0.36$$

e. Refer to the following information and your answer in b. Do you think the turning point is relevant? Explain.

d?

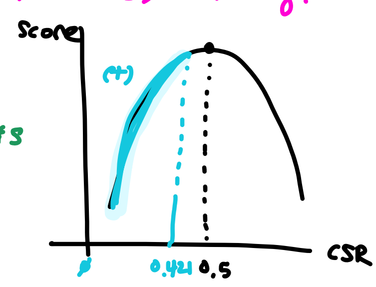
```
. su comp_stu
```

Variable	Obs	Mean	Std. dev.	Min	Max
comp_stu	420	.1359266	.0649558	0	.4208333

Yes. For school districts w/ a CSR below the turning pt (0.36), adding an add'l comp per student increases test scores, on avg.
However, for districts w/ a CSR above the turning pt (btw 0.36 & 0.421), adding one more comp per student decreases test scores, on avg.

What if the turning pt were 0.5?

⇒ No. The turning pt is NOT relevant bcz there are no districts w/ a CSR above 0.5. Therefore, in practice, an add'l comp per student only has a positive effect on test scores.



f. Conduct a significance test at the 5% level to evaluate whether the relationship between the computer-student ratio and test scores is linear. You may use the information in the Stata output.

$$Score_i = \beta_0 + \beta_1 CSR_i + \beta_2 CSR_i^2 + \beta_3 STR_i + u_i$$

① $H_0: \beta_2 = 0 \equiv \beta_{2,0}$ ← (linear)

$H_a: \beta_2 \neq 0$ ← (non linear)

② sig lv $\alpha = 0.05$

③ test statistic $= \dfrac{\hat{\beta}_2 - \beta_{2,0}}{se(\hat{\beta}_2)} = \dfrac{\hat{\beta}_2}{se(\hat{\beta}_2)} = \boxed{-1.32}$

Stata table

④ P-value $= \Pr(|t| > |\text{test statistic}|)$

$= \Pr(|t| > 1.32)$

$= \boxed{0.188}$

⑤ conclusion

p-value vs α

$0.188 > 0.05 \Rightarrow$ We fail to reject Ho that CSR and scores have "linear" relationship.

Appendix: Let's practice interpreting Stata's regression results

We run the following regression model using school district-level data:

$\boldsymbol{Testscore_i = \beta_0 + \beta_1 CompStudent_i + \beta_2 CompStudent_i^2 + \beta_3 STR_i + u_i}$, where i represents the school district,

testscr: Average test scores

str: Student-Teacher Ratio

comp_stu: Computer-Student Ratio

comp_stu2: Computer-Student Ratio squared

testscr	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
comp_stu	123.1777	46.28552	2.66	0.008	32.19501	214.1603
comp_stu2	-173.3259	131.4739	-1.32	0.188	-431.7619	85.11018
str	-1.533944	.4943405	-3.10	0.002	-2.505661	-.5622273
_cons	671.4727	11.16598	60.14	0.000	649.5239	693.4214

Q1. Fill out the table

	Coefficient	Estimated Coefficient	Standard error of estimated coefficient	Test statistic (t-statistic) of the estimated coefficient	p-value of the estimated coefficient	A 95% CI for the parameter [LB, UB]
CompStudent	β_1	$\widehat{\beta_1} =$	$se(\widehat{\beta_1}) =$			[,]
STR	β_3					
Constant (y-intercept)						

Note. CI= Confidence Interval; LB=Lower Bound; UB=Upper Bound

Q2. Calculate the test statistic for $\widehat{\beta_3}$ manually and compare it with the test statistic shown in the Stata table.

Q3. Is the estimated coefficient on *STR* statistically significant at the 5% level?

ANSWERS

Q1.

	Coefficient (Parameter)	Estimated Coefficient	SE of estimated coeff	Test statistic (t-statistic) of the estimated coefficient	p-value of the estimated coefficient	A 95% CI for the parameter [LB, UB]
CompStudent	β_1	$\widehat{\beta_1} = 123.1777$	$se(\widehat{\beta_1}) = 46.28552$	2.66	0.008	[32.19501 , 214.1603]
STR	β_3	$\widehat{\beta_3} = -1.533944$	$se(\widehat{\beta_3}) = .4943405$	-3.10	0.002	[-2.505661 , -.5622273]
Constant (y-intercept)	β_0	$\widehat{\beta_0} = 671.4727$	$se(\widehat{\beta_0}) = 11.16598$	60.14	0.000	[649.5239 , 693.4214]

Q2. Test-statistic $= \frac{\widehat{\beta_3}}{se(\widehat{\beta_3})} = \frac{-1.533944}{.4943405} = -3.10$

Q3. Yes. The p-value of $\widehat{\beta_3}$ is 0.002, which is smaller than 0.05. Therefore, the estimated coefficient is statistically significant at the 5% level. In other words, we reject the null hypothesis that $\beta_3 = 0$, implying that there is a relationship between the ~~computer-student ratio~~ STR and test scores.